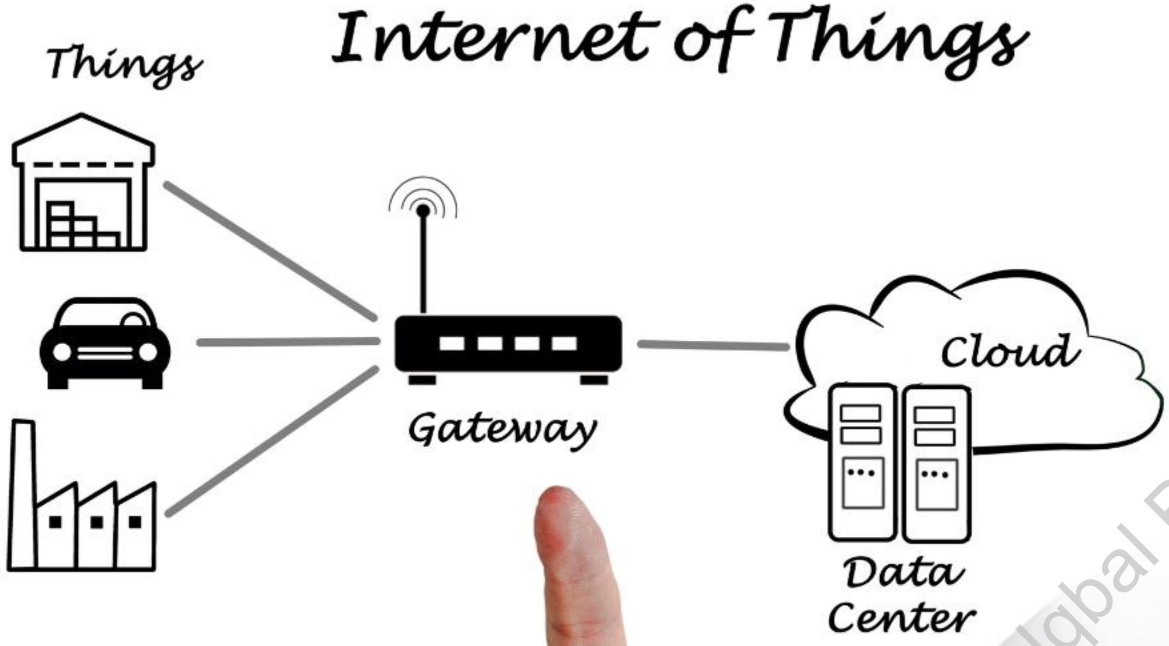
Architecture of

Internet of Things



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What is IoT Architecture?

- **IoT architecture refers to the multi-layered framework that enables the interconnectedness of smart devices, sensors, and software.**
- **This network consists of smart sensors, actuators and other connected elements that enable data flow from physical sources through networks into storage in the cloud thereby allowing for intelligent decision-making and automation.**

Need of IoT Architecture?

EFFICIENCY: STREAMLINES THE MANAGEMENT OF VAST NETWORKS OF IOT DEVICES.

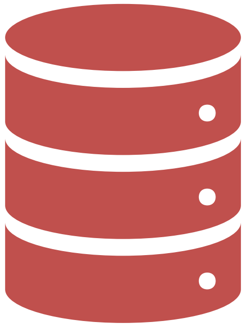
SCALABILITY: FACILITATES THE GROWTH OF IOT NETWORKS AS DEMAND INCREASES.

SECURITY: ENSURES DATA INTEGRITY AND PROTECTION ACROSS THE NETWORK.

INTEROPERABILITY: ALLOWS DIVERSE DEVICES AND SYSTEMS TO COMMUNICATE SEAMLESSLY.

INNOVATION: ENABLES THE DEVELOPMENT OF NEW APPLICATIONS AND SERVICES.

Basic IoT Architectures

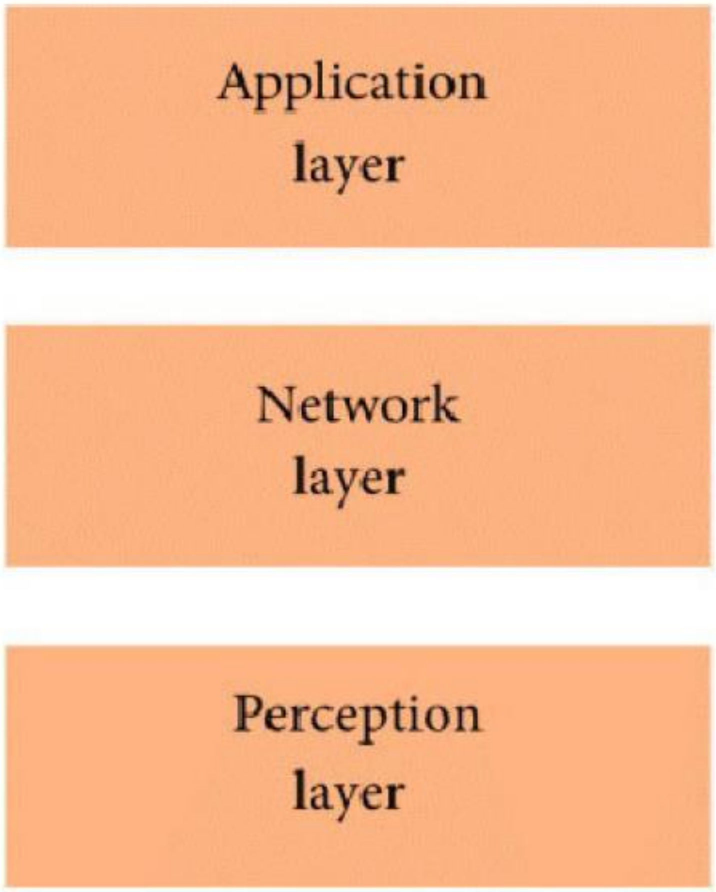


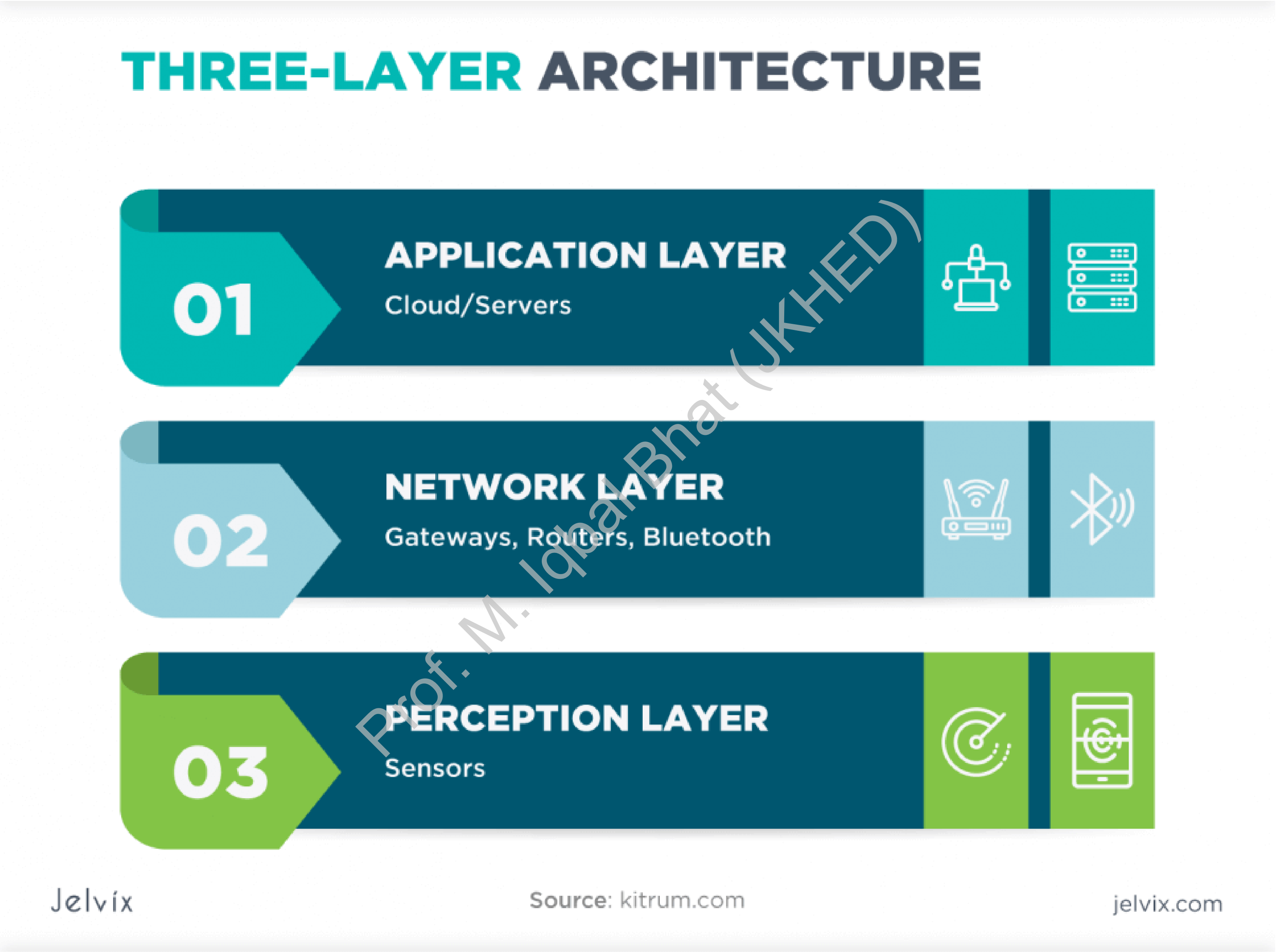
3 layer Architecture

5-layer Architecture

3-Layer IoT Architecture

- This is the first and most basic three-layer architecture is
- It was established in the initial phases of study in this area.
- The architecture comprises of three layers, namely:
  - – perception,
  - – network, and
  - – application layers




THREE-LAYER ARCHITECTURE
01
APPLICATION LAYER
Cloud/Servers
02
NETWORK LAYER
Gateways, Routers, Bluetooth
03
PERCEPTION LAYER
Sensors
Jelvix
Source: kitrum.com
jelvix.com

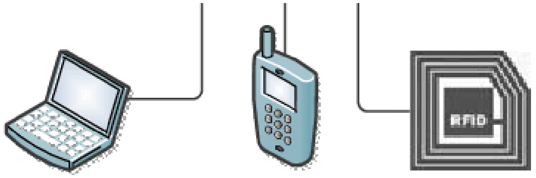
3-Layer IoT Architecture

Perception Layer

The foundation of IoT, consisting of sensors and devices.

Collects and provides raw data from the environment.

Examples: Temperature sensors, motion detectors.

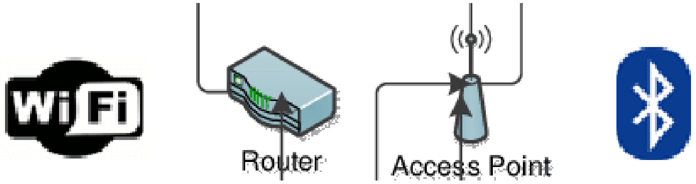


Network Layer

The bridge between the perception and application layers.

Transmits the collected data to processing systems or the cloud.

Utilizes technologies like Wi-Fi, Bluetooth, and cellular networks.

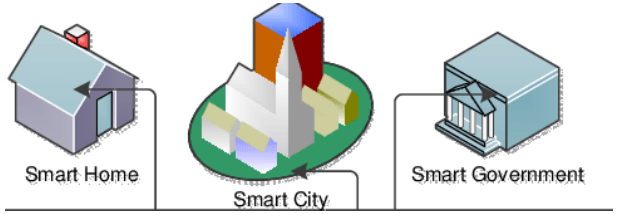


Application Layer

Delivers the final service to the user based on processed data.

Enables user interaction and control over IoT devices.

Examples: Smart home apps, health monitoring systems.



Significance of 3-Layer Architecture

Simplicity: Provides a straightforward model for IoT connectivity.

Flexibility: Allows for various technologies at each layer.

Scalability: Supports the addition of more devices and services.

Need to High Layer Architectures

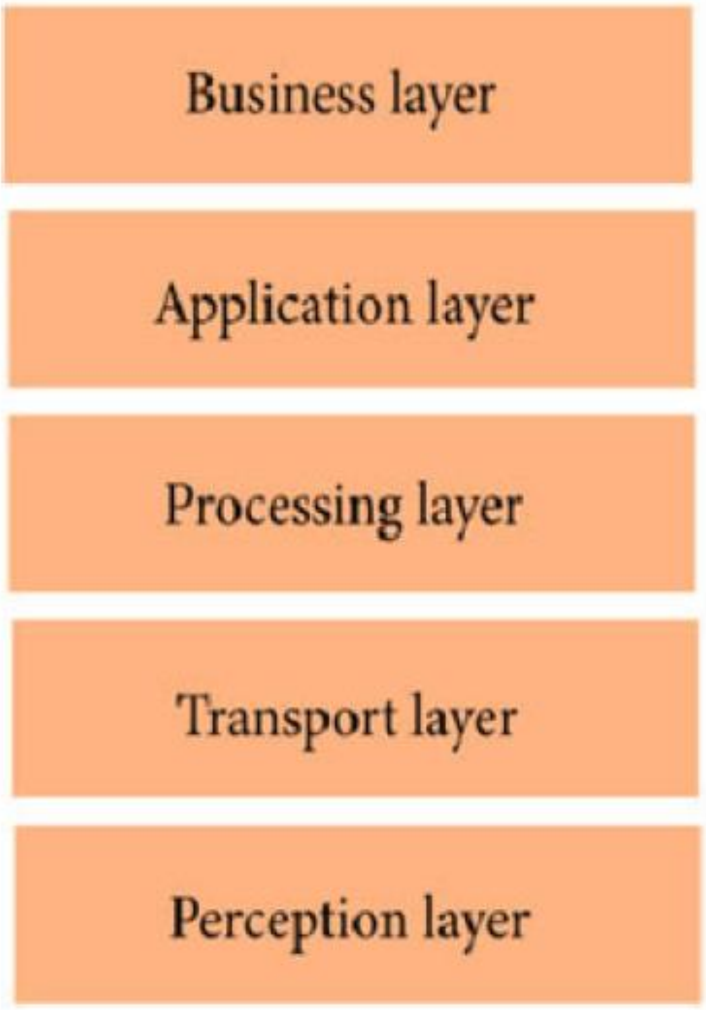
The three-layer architecture defines the main idea of the IoT, but it is not sufficient for research on IoT because research often focuses on finer aspects of the IoT.

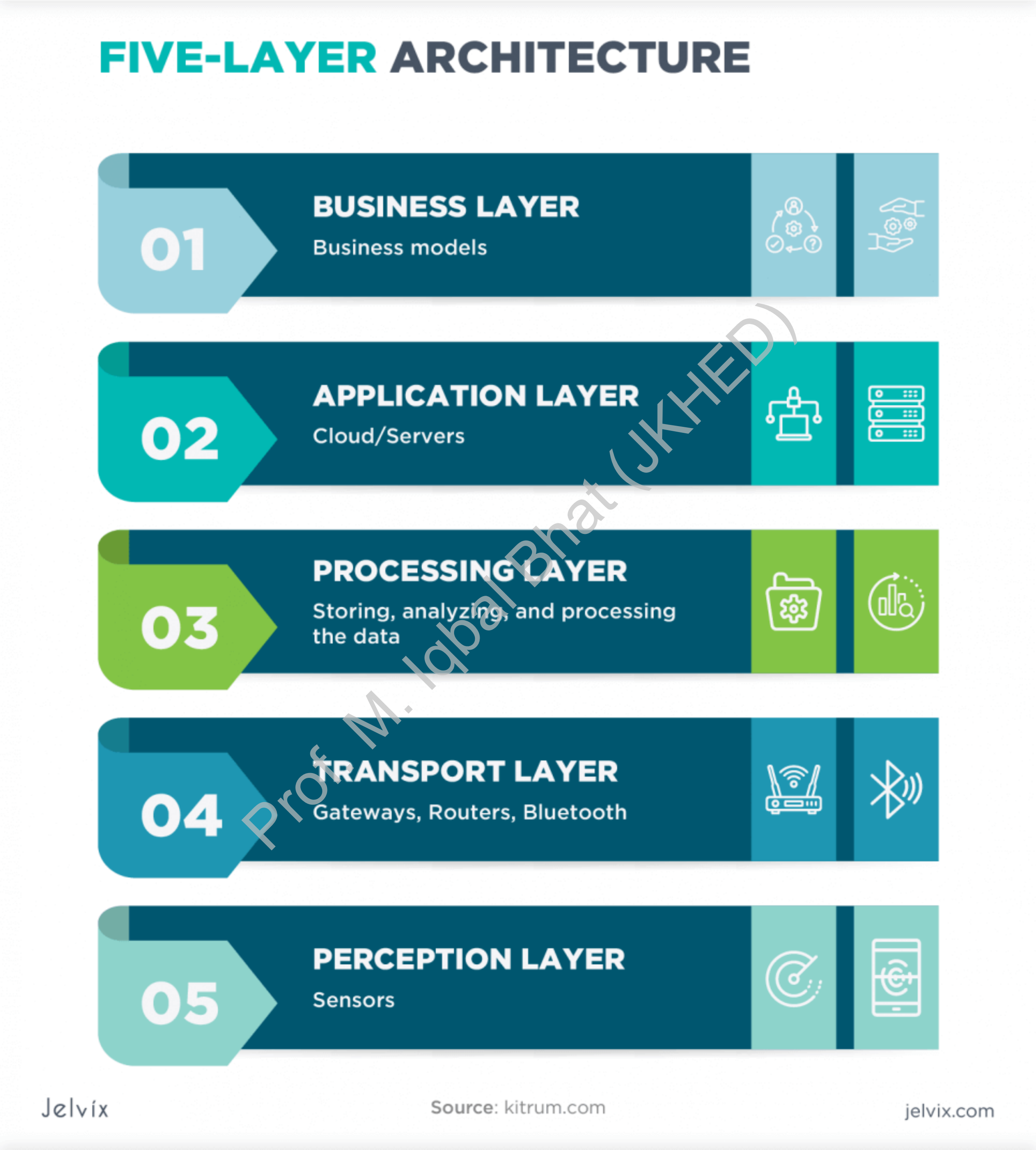
There is a need to manage more complex systems and provide advanced functionalities.

That is why we have many more layered architectures like 5-layer IoT Architecture.

5-Layer IoT Architecture

- The 5-layer IoT architecture is a conceptual framework that breaks down the components and functions of an IoT system into distinct layers.
- This layered approach helps to understand the complex interactions within an IoT ecosystem and facilitates the design, development, and deployment of efficient and scalable solutions.
- The architecture comprises of three layers, namely:
  - perception,
  - Transport layer
  - Processing layer
  - Application Layer
  - Business Layer




FIVE-LAYER ARCHITECTURE
01
BUSINESS LAYER
Business models
02
APPLICATION LAYER
Cloud/Servers
03
PROCESSING LAYER
Storing, analyzing, and processing the data
04
TRANSPORT LAYER
Gateways, Routers, Bluetooth
05
PERCEPTION LAYER
Sensors
Jelvix
Source: kitrum.com
jelvix.com

5-Layer IoT Architecture

Perception Layer:

The sensory interface to the physical world.

Collects data through sensors and actuators.

Examples: Thermometers, cameras, microphones.

Transport Layer:

The transport layer is responsible for transferring sensor's data from the perception layer to the processing layer and vice versa by means of a variety of connectivity such as RFID, NFC, LAN, Bluetooth, and many more.

Processing Layer:

The processing layer is the brain of the IoT ecosystem.

Typically, data is analyzed, pre-processed, and stored here before being sent to the data center, where it is accessed by software applications that both monitor and manage the data as well as prepare further actions

Application Layer:

Delivers specific applications to end-users.

Interfaces for user interaction and control.

Examples: Smart home controls, health monitoring apps.

Business Layer:

Manages the overall business model. Analyzes data for business insights and decisions.

This layer uses the data that are received from the network layer to build various components such as business models, graphs, and flowcharts. Examples: User behavior analytics, service deployment strategies

Benefits of 5-layer architecture

Complexity Management: Better handles the complexity of large-scale IoT systems.

Enhanced Functionality: Provides advanced data processing and analytics.

Improved Security: Offers more robust security features across layers.

Business Integration: Aligns IoT operations with business objectives.

Questions?